

Daniele Alessi

Contacts

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Personal Website & Portfolio:

<https://eldanee.github.io>

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Education

Master degree in Computer Science Engineering @ Politecnico di Milano
(September 2021 - April 2025)

Erasmus Programme @ IST Universidade de Lisboa
(February 2023 - July 2023)

Bachelor degree in Computer Science Engineering @ Politecnico di Milano
(September 2018 - September 2021)

Experience

Graphics & Rendering Junior Programmer @ Untold Games; Genoa, Italy
(June 2025 - Present)

IT Specialist Intern @ Bending Spoons; Milan, Italy
(March 2019 - September 2020)

Skills

Computer Graphics

- Theoretical and practical knowledge of rendering pipelines and shading techniques.
- Experience with CPU and GPU accelerated ray tracing.
- Hands-on experience with various culling techniques.
- Experience working with Vulkan and GLSL, and basic knowledge of a few other graphics API and shading languages.
- Familiar with Unreal Engine, and junior experience adding and adapting features to the engine codebase.

GPU programming

- Theoretical knowledge of GPU architectures and kernel execution.
- Realized a GPU accelerated Position Based Dynamics simulation using Vulkan Compute for my master thesis.
- GPU profiling using RenderDoc, Pix and various platform specific tools.

Video Game Design and Programming

- Designed and programmed a video game in Unity for an academic project.
(Beta version available at <https://polimi-game-collective.itch.io/diagonaliz>)
- Smaller projects submitted to various Game Jams.

Other

Languages

Italian, English

Best Paper Award 2025 @ ASME VARE

Received the ASME Virtual and Augmented Reality Environments Best Paper Award 2025 at the IDETC/CIE 2025, presenting the paper "*Efficient Soft Body Simulation for Real-Time Applications: A GPU-Based XPBD Approach*", Alessi, D, Buttiglione, MD, Lucania, E, Piazzolla, P, Colombo, G, Gribaudo, M.
The abstract can be found [here](#).